

- 5 A block of mass 27 g is suspended from a spring so that it rests in equilibrium, as shown in Figure 5.1.

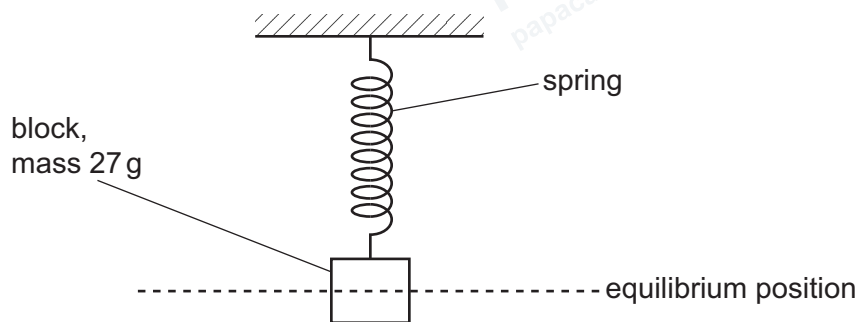


Figure 5.1

The block is displaced downwards through a small distance. The block is released and then oscillates vertically with simple harmonic motion.

The oscillations have a period of 1.7 s and an energy of $13\text{ }\mu\text{J}$.

- (a) (i) Show that the angular frequency of the oscillations is 3.7 rad s^{-1} .

[1]

- (ii) State what is meant by the amplitude of the oscillations.

.....

.....

..... [1]

- (iii) Calculate the amplitude of the oscillations.

amplitude = m [3]

- (b) On Figure 5.2, sketch the variation of the kinetic energy E_K of the block with displacement x from the equilibrium position for one complete oscillation of the block. Add a suitable scale to the x -axis.

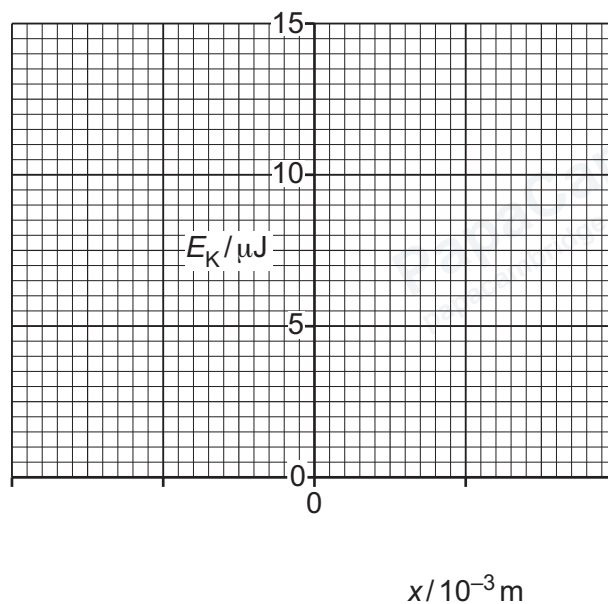


Figure 5.2

[3]

[Total: 8]